



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO JEYAMMAL TRADERS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

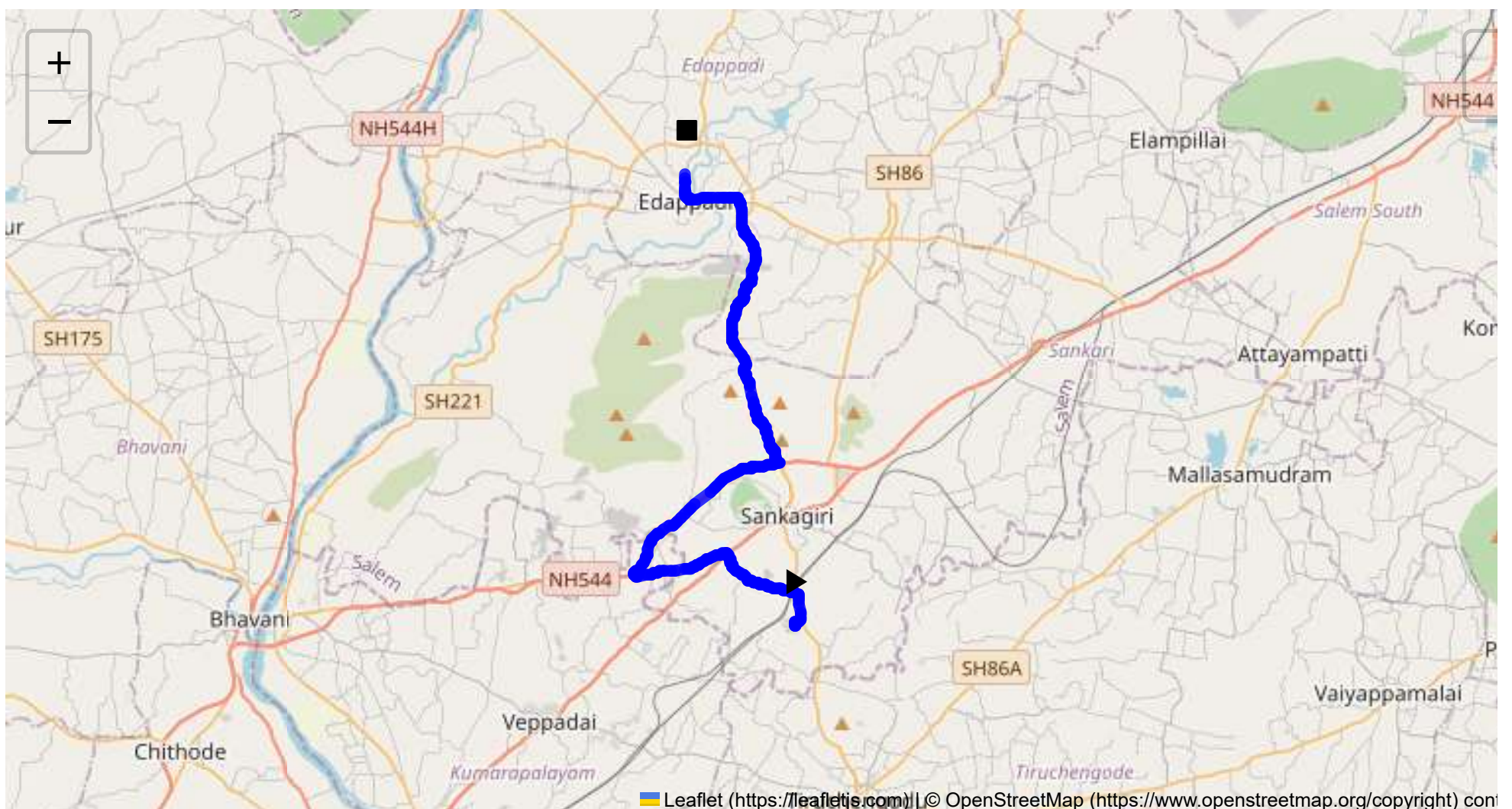
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 29.76 km**  
**Estimated Duration: 0.7 hours**  
**Adjusted Duration (Heavy Vehicle): 0.9 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.592872, 77.83469)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from Sangagiri to Edappadi via the specified points in Tamil Nadu covers approximately 29.76 kilometers and includes several intersecting roads. It progresses primarily through regional areas with a focus on highway access and smaller connecting roads, which include Bhavani Main Road and the

Sankari-Edappadi-Mecheri Road. The route is characterized by a combination of rural landscapes and small townships.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate:

- **Summer (March to June):** High temperatures can reach up to 40°C (104°F), risking heat-related mechanical issues.
- **Monsoon (June to September):** Heavy rains can lead to waterlogged roads and reduced visibility. Flash flooding is possible in low-lying areas.
- **Post-Monsoon (October to November):** Continued rain can affect road conditions; wet and slippery surfaces need careful navigation.
- **Winter (December to February):** Cooler temperatures with clear skies, favorable for driving.

## 3. Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Morning (7 AM - 10 AM) and evening (5 PM - 8 PM) see increased traffic from local commuters.
- **Congestion Areas:** Town centers along the route, particularly near markets and school zones, are prone to congestion.

## 4. Assessment of Road Quality and Infrastructure

- **Main Roads:** Bhavani Main Road and Sankari-Edappadi-Mecheri Road are generally well-maintained with clear signage.
- **Rural Segments:** Watch for narrower lanes and occasional potholes, which can be hazardous, particularly during monsoons.

## 5. Suggestions for Alternative Routes for Emergencies

- **Via NH 544 & NH 79:** These national highways provide a more secure route during major blockages or flooding.
- **Through Salem Utility Roads:** Allows bypass of congested areas in emergency situations.

## 6. Local Regulations Affecting Hazardous Material Transport

Local regulations restrict the movement of hazardous materials, especially through densely populated regions during peak times and festivals. Ensure compliance with signage for hazardous materials and secure proper permits for transport.

## 7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Common Incidents:** Mostly involve overloading and lack of compliance with safety norms.



- **Noteworthy:** Rare incidents involving spillage have highlighted the need for caution at key junctions and turns.

## 8. Environmental Considerations and Sensitive Areas

- **Water bodies:** Nearby lakes and rivers necessitate careful handling to prevent contamination.
- **Wildlife Crossings:** Be cautious of areas marked for animal crossings, especially at night.

## 9. Analysis of Communication Coverage

- **Good Coverage:** Major towns and highways.
- **Potential Dead Zones:** Ridge areas or deep rural zones may experience coverage issues, especially under dense vegetation.

## 10. Estimated Emergency Response Times

- **Urban Response (Salem):** Approximately 15-25 minutes.
- **Rural Response:** Could range from 30 minutes to over an hour; coordination with local services is advised.

## 12. Overall Summary of Risk Assessment

The route is moderately risky for heavy vehicles, particularly those carrying hazardous materials, due to road conditions in rural areas and weather impacts. The infrastructure supports safe transit but requires cautious navigation around populated and environmentally sensitive areas. Reliable communication is a key factor, with potential challenges in more remote sections. Adequate spill control measures and emergency procedures must be established, considering reduced response times in rural segments.

Regular updates on weather and communication with local traffic control can further mitigate risks. Utilization of alternative routes in emergencies, alongside adherence to local regulations, ensures a safer transit pathway for hazardous materials.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.58435, 77.83651 | 15 KM/Hr    |
| 1 | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 2 | Turn       | Medium     | 11.43956, 77.87340 | 30 KM/Hr    |
| 3 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 4 | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 5 | Turn       | Medium     | 11.44898, 77.87410 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 6  | Turn       | Medium     | 11.45357, 77.85789 | 30 KM/Hr    |
| 7  | Turn       | Medium     | 11.45348, 77.85706 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.45352, 77.85698 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.46318, 77.84913 | 30 KM/Hr    |
| 10 | Turn       | High       | 11.45542, 77.81725 | 15 KM/Hr    |
| 11 | Turn       | High       | 11.45631, 77.81668 | 15 KM/Hr    |
| 12 | Turn       | Medium     | 11.49249, 77.85821 | 30 KM/Hr    |
| 13 | Blind Spot | Blind Spot | 11.49390, 77.86748 | 10 KM/Hr    |
| 14 | Turn       | Medium     | 11.58296, 77.85419 | 30 KM/Hr    |
| 15 | Turn       | High       | 11.58523, 77.83507 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                             | coordinates            | speed_limit | risk_level |
|----|----------|----------------------------------|------------------------|-------------|------------|
| 1  | hospital | KRS Hospital and Diabetic Centre | 11.5848828, 77.8449664 | 30 km/h     | Medium     |
| 2  | hospital | Kannan Hospital                  | 11.5854802, 77.8425936 | 30 km/h     | Medium     |
| 3  | hospital | Aravind Hospital                 | 11.5854615, 77.8424165 | 30 km/h     | Medium     |
| 4  | hospital | Muthu Eyecare Hospital           | 11.584736, 77.8393794  | 30 km/h     | Medium     |
| 5  | clinic   | Uma Clinic                       | 11.58398, 77.83977     | 30 km/h     | Medium     |
| 6  | clinic   | KGP Clinic                       | 11.5847763, 77.8387488 | 30 km/h     | Medium     |
| 7  | hospital | Government Hospital, Edapadi     | 11.5848264, 77.8380174 | 30 km/h     | Medium     |
| 8  | hospital | Sri Sankar Hospital              | 11.5862984, 77.8355021 | 30 km/h     | Medium     |
| 9  | hospital | Sri Advika Hospital              | 11.585049, 77.834422   | 30 km/h     | Medium     |
| 10 | clinic   | Raj Physiotherapy Clinic         | 11.5857357, 77.8341218 | 30 km/h     | Medium     |
| 11 | hospital | Dr. O.S. Subramanium Hospital    | 11.584996, 77.834235   | 30 km/h     | Medium     |
| 12 | hospital | Aravinthu Eye Hospital           | 11.5866631, 77.8341465 | 30 km/h     | Medium     |
| 13 | clinic   | Xcellent Homecare Clinic         | 11.584762, 77.8338085  | 30 km/h     | Medium     |
| 14 | hospital | Mohan Hospital                   | 11.5841331, 77.8336277 | 30 km/h     | Medium     |
| 15 | hospital | SKM Hospital                     | 11.5841611, 77.8334397 | 30 km/h     | Medium     |
| 16 | hospital | Nivetha Hospital                 | 11.5858509, 77.833451  | 30 km/h     | Medium     |
| 17 | clinic   | Krishna Hospital and Clinic      | 11.5846542, 77.8333239 | 30 km/h     | Medium     |
| 18 | hospital | Rathanasamy Hospital             | 11.58583, 77.833443    | 30 km/h     | Medium     |

|    | type     | name               | coordinates            | speed_limit | risk_level |
|----|----------|--------------------|------------------------|-------------|------------|
| 19 | hospital | Pachiappa Hospital | 11.58571, 77.833131    | 30 km/h     | Medium     |
| 20 | hospital | Arunagiri Hospital | 11.5837095, 77.8332129 | 30 km/h     | Medium     |

## Crowded Spots

|   | type   | name                       | coordinates            | speed_limit | risk_level |
|---|--------|----------------------------|------------------------|-------------|------------|
| 0 | school | KRP Matric. Hr. Sec School | 11.4546193, 77.8142445 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.58435, 77.83651





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44898, 77.87410



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45357, 77.85789





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45348, 77.85706



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45352, 77.85698





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46318, 77.84913



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45542, 77.81725



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45631, 77.81668





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49249, 77.85821



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.49390, 77.86748





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.58296, 77.85419



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.58523, 77.83507